Nama : Alif Mahtum Alfaidz

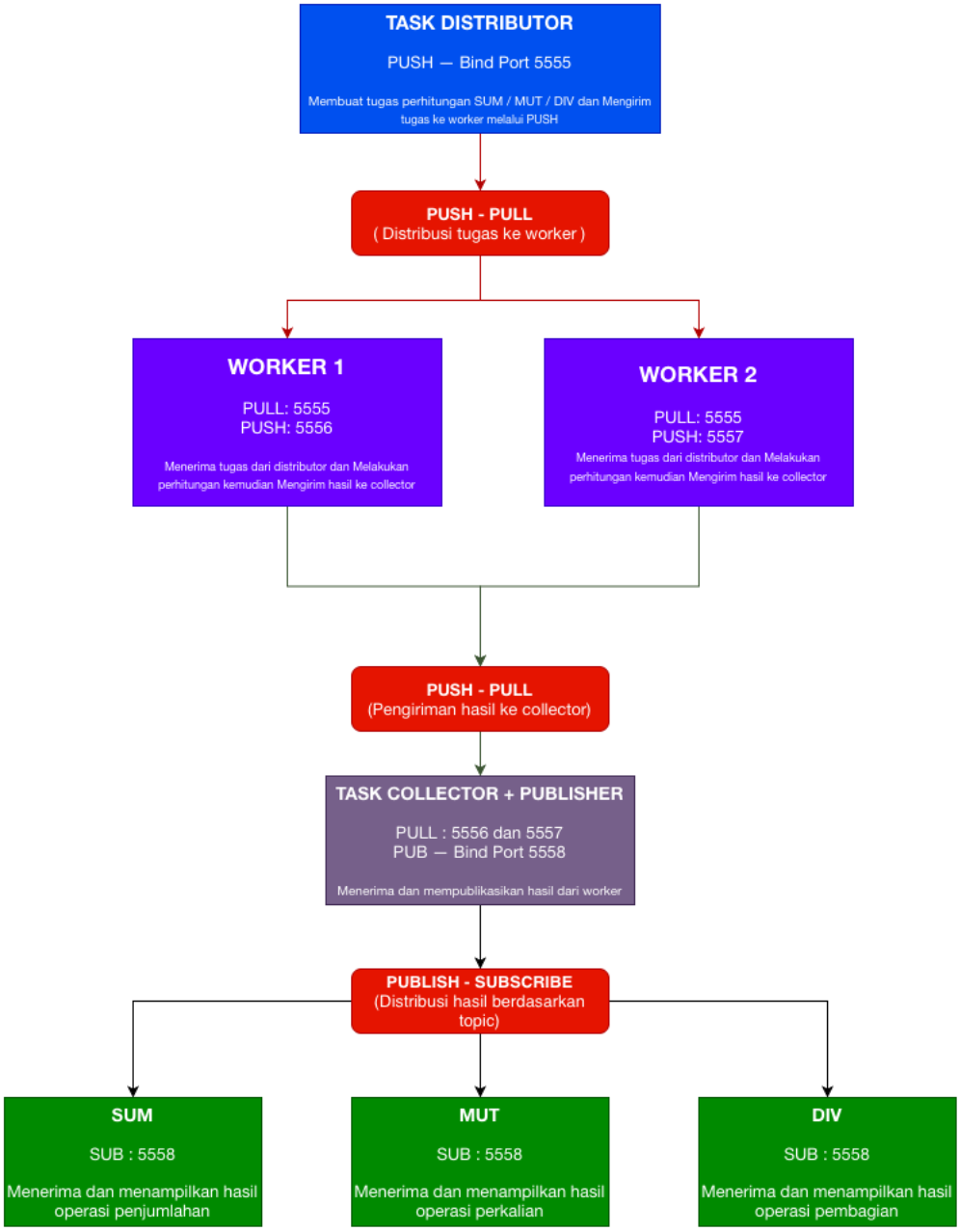
NIM : 103012580030

TASK C — COMBINATION OF PUSH-PULL AND PUBLISH-SUBSCRIBE PATTERNS

A. Deskripsi Progam

Pada Task ini saya membuat program komunikasi sistem terdistribusi menggunakan bahasa pemrograman Python dengan library PyZMQ (ZeroMQ). Pada program ini menggabungkan dua pola komunikasi, yaitu Push-Pull dan Publish-Subscribe. untuk mengirim tugas perhitungan dan menampilkan hasilnya. Program terdiri dari satu Distributor, dua Worker dan satu Collector yang juga berfungsi sebagai Publisher serta tiga Subscriber. Tugas yang dikerjakan berupa operasi matematika sederhana yaitu penjumlahan (SUM), perkalian (MUT), dan pembagian (DIV). Pada pola Push-Pull, Distributor mengirimkan tugas berupa dua angka acak dan jenis operasi kepada Worker 1 atau Worker 2. Worker yang menerima tugas selanjutnya akan melakukan perhitungan kemudian mengirimkan hasilnya ke Collector. Selanjutnya Collector tersebut akan menerima hasil perhitungan dan menampilkan menggunakan pola Publish-Subscribe. Setiap Subscriber hanya menerima hasil sesuai dengan topic yang sudah ditentukan, yaitu SUM, MUT, atau DIV. Program juga dijalankan menggunakan Docker Compose agar setiap komponen dapat berjalan dalam container yang berbeda dan tetap bisa saling berkomunikasi.

B. Ilustrasi Arsitektur Sistem



Berdasarkan ilustrasi pada Gambar, sistem menggunakan kombinasi pola komunikasi Push-Pull dan Publish-Subscribe untuk membagikan tugas perhitungan serta mengirimkan hasilnya kepada Subscriber. Alur komunikasi pada sistem ini terdiri dari beberapa tahapan sebagai berikut.

1. **Task Distributor**

   Proses dimulai dari Task Distributor yang membuat dua angka secara acak dan menentukan salah satu operasi matematika, yaitu penjumlahan (SUM), perkalian (MUT), atau pembagian (DIV). Selanjutnya, Distributor mengirimkan tugas tersebut menggunakan socket PUSH melalui port 5555. Pengiriman tugas dilakukan secara berkala dengan jeda 5 detik.

2. **Worker 1 dan Worker 2**

   Worker 1 dan Worker 2 menerima tugas dari Distributor menggunakan socket PULL yang terhubung ke port 5555. Setiap tugas akan diterima oleh salah satu Worker yang tersedia, sehingga kedua Worker dapat berbagi pekerjaan. Setelah menerima tugas, Worker melakukan perhitungan sesuai jenis operasi yang diberikan. Hasil perhitungan kemudian dikirimkan ke Task Collector menggunakan socket PUSH, yaitu melalui port 5556 untuk Worker 1 dan port 5557 untuk Worker 2.

3. **Task Collector + Publisher**

   Task Collector menerima hasil perhitungan dari kedua Worker melalui socket PULL yang terhubung ke port 5556 dan 5557. Collector menggunakan Poller untuk memantau pesan yang masuk dari kedua Worker. Setelah hasil diterima, Collector menambahkan pesan status berdasarkan jenis operasi dan menampilkan informasi hasil perhitungan. Selanjutnya, Collector yang juga berperan sebagai Publisher mempublikasikan hasil tersebut melalui socket PUB pada port 5558.

4. **Subscriber**

   Pada tahap terakhir, terdapat tiga Subscriber, yaitu SUM, MUT, dan DIV, yang menggunakan socket SUB untuk menerima pesan dari Publisher melalui port 5558. Setiap Subscriber hanya menerima pesan sesuai topic yang didaftarkan. SUM Subscriber menerima hasil penjumlahan, MUT Subscriber menerima hasil perkalian, dan DIV Subscriber menerima hasil pembagian. Pesan yang diterima kemudian ditampilkan pada terminal, meliputi angka yang dihitung, jenis operasi, hasil perhitungan, serta Worker yang mengerjakan tugas tersebut.

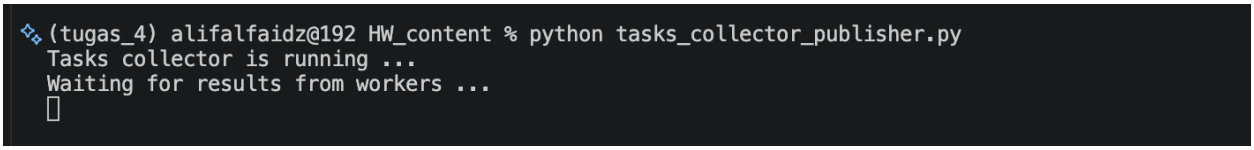
Dengan alur tersebut, pola Push-Pull digunakan untuk membagi dan memproses tugas melalui Worker, sedangkan pola Publish-Subscribe digunakan untuk menyampaikan hasil perhitungan kepada Subscriber berdasarkan topic masing-masing. Seluruh komponen juga berhasil dijalankan menggunakan Docker Compose sehingga dapat saling berkomunikasi melalui jaringan antar-container.

C. Hasil Implementasi Dan Pengujian

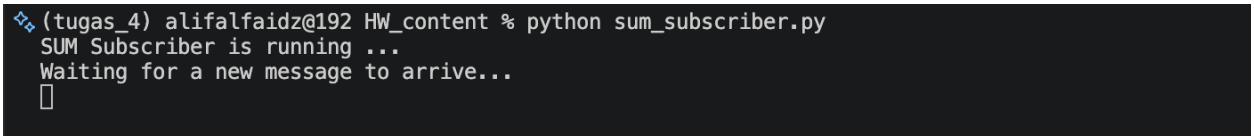
Setelah seluruh kode program selesai dibuat dilakukan pengujian untuk memastikan setiap komponen dapat berjalan dan saling berkomunikasi dengan baik. Pengujian dilakukan menggunakan Python dengan library PyZMQ melalui terminal Visual Studio Code.

Selain itu program juga dijalankan menggunakan Docker Compose untuk memastikan komunikasi antar-container dapat berjalan. Hasil pengujian menunjukkan bahwa Distributor berhasil mengirimkan tugas dan kedua Worker dapat melakukan perhitungan. Collector berhasil menerima serta mempublikasikan hasil kepada Subscriber berdasarkan topic masing-masing. Berikut merupakan hasil pengujian dari setiap komponen program.

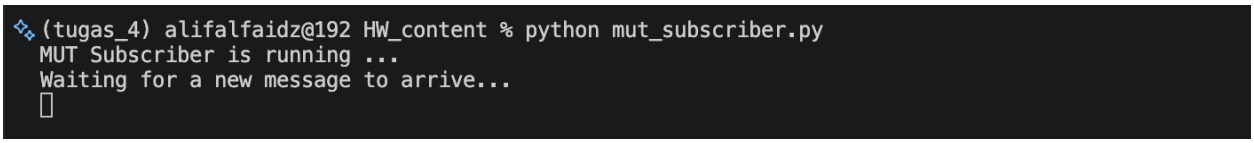
A. Terminal Collector + Publisher



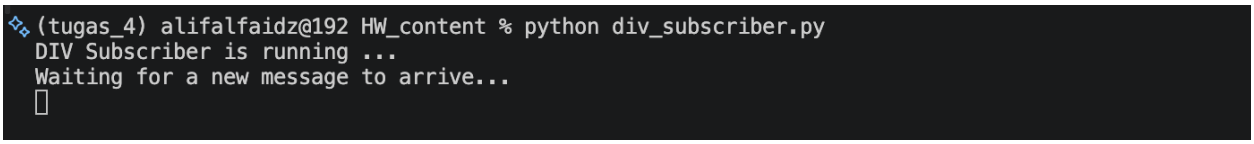
B. Terminal SUM Subscriber



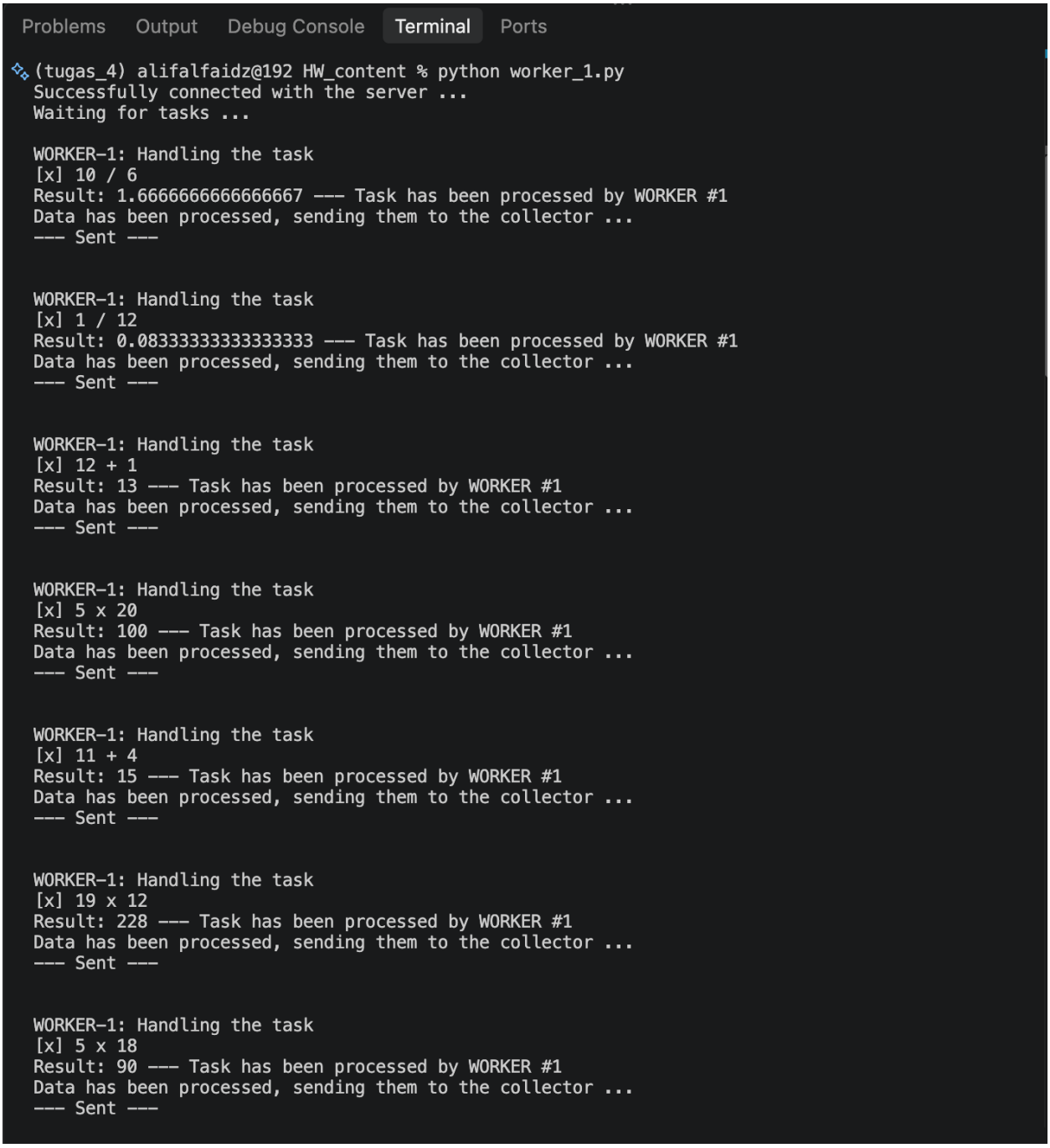
C. Terminal MUT Subscriber



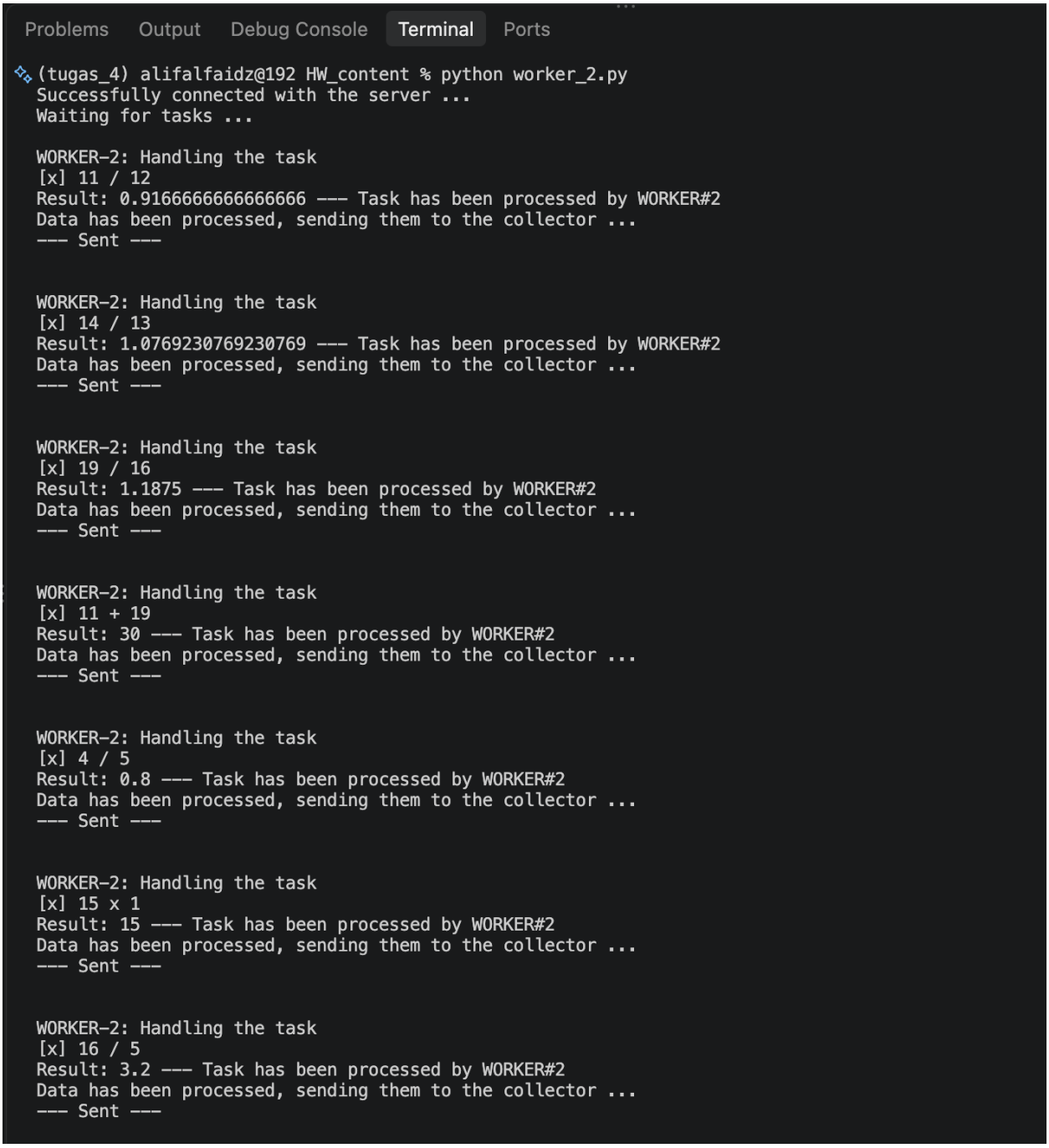
D. Terminal DIV Subscriber



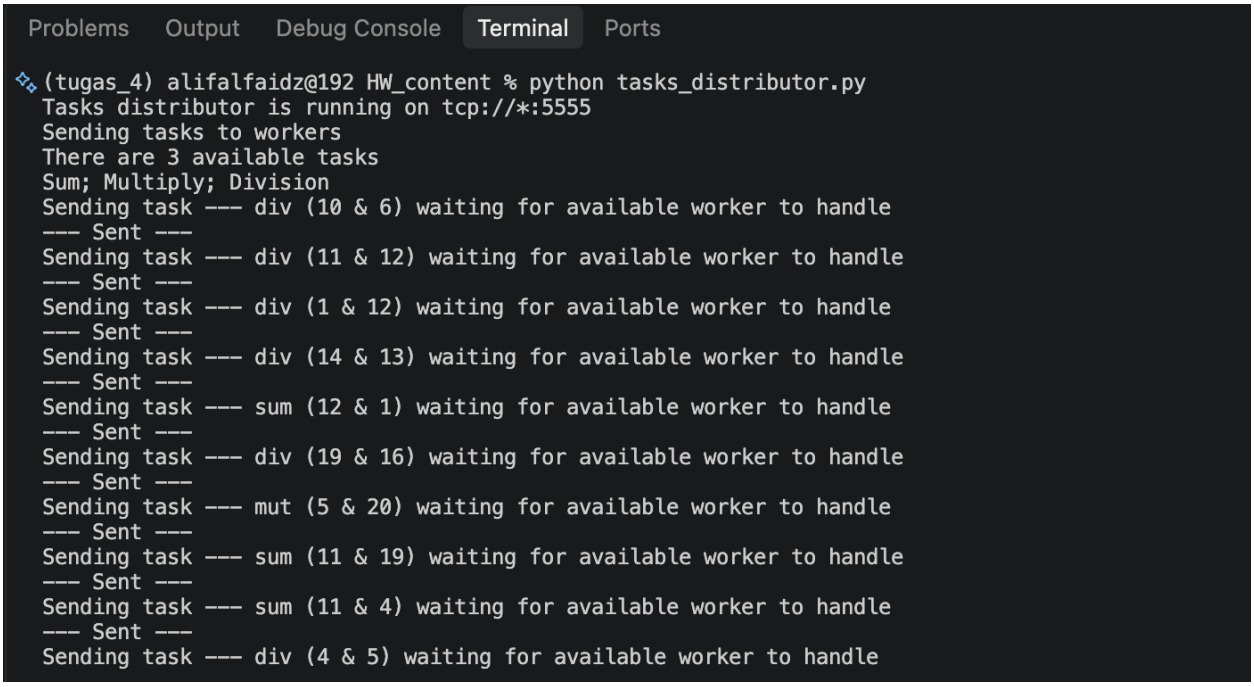
E. Terminal Worker 1



F. Terminal Worker 2



G. Terminal Distributor



H. Docker

```
(tugas_4) alifalfaidz@192 HW_content % docker compose up -d --build
[+] Building 58.2s (26/26) FINISHED
 => [internal] load local bake definitions                          0.0s
 => => reading from stdin 4.60kB                                    0.0s
 => [mut_subscriber internal] load build definition from Docke      0.0s
 => => transferring dockerfile: 1.30kB                              0.0s
 => [distributor internal] load metadata for docker.io/library      1.3s
 => [auth] library/python:pull token for registry-1.docker.io       0.0s
 => [sum_subscriber internal] load .dockerignore                    0.0s
 => => transferring context: 105B                                   0.0s
 => [div_subscriber 1/6] FROM docker.io/library/python:3.9-sli      0.0s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6      0.0s
 => [worker2 internal] load build context                           0.0s
 => => transferring context: 3.49kB                                 0.0s
 => CACHED [collector 2/6] WORKDIR /app                             0.0s
 => [sum_subscriber 3/6] COPY . .                                   0.0s
 => [worker2 4/6] RUN python -m pip install --no-cache-dir py      54.7s
 => [mut_subscriber 5/6] RUN sed -i 's|tcp://localhost:5555|tc      0.1s
 => [worker1 6/6] RUN grep -q 'tcp://distributor:5555' worker_      0.1s
 => [mut_subscriber] exporting to image                             1.6s
 => => exporting layers                                             1.0s
 => => exporting manifest sha256:343ebcb4663530cfc24f5989d15d3      0.0s
 => => exporting config sha256:e46a08dff1b46587e7a55d2eac4e593      0.0s
 => => exporting attestation manifest sha256:3ccf80e4b61cb26c3      0.0s
 => => exporting manifest list sha256:35f126cbd5a3c8c249369dcd      0.0s
 => => naming to docker.io/library/hw_content-mut_subscriber:l      0.0s
 => => unpacking to docker.io/library/hw_content-mut_subscribe      0.5s
 => [div_subscriber] exporting to image                             1.7s
 => => exporting layers                                             1.0s
 => => exporting manifest sha256:3d1b0667795f532b4c8c457119609      0.0s
 => => exporting config sha256:284eaeca10215e07bfd8f95e34c56d0      0.0s
 => => exporting attestation manifest sha256:0050e6c1609e6967a      0.0s
 => => exporting manifest list sha256:857c4617c7f9b5826e366bd9      0.0s
 => => naming to docker.io/library/hw_content-div_subscriber:l      0.0s
 => => unpacking to docker.io/library/hw_content-div_subscribe      0.6s
 => [distributor] exporting to image                                1.7s
 => => exporting layers                                             1.0s
 => => exporting manifest sha256:9bc6fe60724bab6afc0cfb4332bef      0.0s
 => => exporting config sha256:339c6c762d11da0ba7e00df05344987      0.0s
 => => exporting attestation manifest sha256:beceee105a87e43b9      0.0s
 => => exporting manifest list sha256:2550bb33620bf06d05450aaa      0.0s
 => => naming to docker.io/library/hw_content-distributor:late      0.0s
 => => unpacking to docker.io/library/hw_content-distributor:l      0.5s
 => [sum_subscriber] exporting to image                             1.6s
 => => exporting layers                                             1.0s
 => => exporting manifest sha256:2b90d02a99db51cf8d0b7c6d4eb11      0.0s
 => => exporting config sha256:e412a614bfe3354cd6dc4da62b530cb      0.0s
 => => exporting attestation manifest sha256:5132d2db27620b684      0.0s
 => => exporting manifest list sha256:75f805d51306e8f6e31ec57b      0.0s
 => => naming to docker.io/library/hw_content-sum_subscriber:l      0.0s
 => => unpacking to docker.io/library/hw_content-sum_subscribe      0.5s
 => [worker2] exporting to image                                    1.6s
```

```
(tugas_4) alifalfaidz@192 HW_content % docker compose up -d --build
 => [worker2] exporting to image                                    1.6s
 => => exporting layers                                             1.0s
 => => exporting manifest sha256:ccc6d7d437528d053e27a7794fc82       0.0s
 => => exporting config sha256:9358910215d7942558c060a9c6a4f0e       0.0s
 => => exporting attestation manifest sha256:4fe480870978dedd5      0.0s
 => => exporting manifest list sha256:0c7c631f285bb7dfe7f64225      0.0s
 => => naming to docker.io/library/hw_content-worker2:latest        0.0s
 => => unpacking to docker.io/library/hw_content-worker2:lates      0.5s
 => [collector] exporting to image                                  1.6s
 => => exporting layers                                             1.0s
 => => exporting manifest sha256:6bf2b7feae3274893532a451c5ac0      0.0s
 => => exporting config sha256:994b375ef793010cf7126cbed7554d8      0.0s
 => => exporting attestation manifest sha256:9cd1341e4a0b1843d      0.0s
 => => exporting manifest list sha256:323775a475a5a9d73c2554ba      0.0s
 => => naming to docker.io/library/hw_content-collector:latest      0.0s
 => => unpacking to docker.io/library/hw_content-collector:lat      0.5s
 => [worker1] exporting to image                                    1.6s
 => => exporting layers                                             1.0s
 => => exporting manifest sha256:01bfde922ef696d98a1a6db3d0670      0.0s
 => => exporting config sha256:86f5d51cbb87573819a050ff3485a5b      0.0s
 => => exporting attestation manifest sha256:11df55064c6a20ab4      0.0s
 => => exporting manifest list sha256:ac006f09e846239d867f8acf      0.0s
 => => naming to docker.io/library/hw_content-worker1:latest        0.0s
 => => unpacking to docker.io/library/hw_content-worker1:lates      0.5s
 => [worker2] resolving provenance for metadata file                0.0s
 => [mut_subscriber] resolving provenance for metadata file         0.0s
 => [worker1] resolving provenance for metadata file                0.0s
 => [div_subscriber] resolving provenance for metadata file         0.0s
 => [collector] resolving provenance for metadata file              0.0s
 => [sum_subscriber] resolving provenance for metadata file         0.0s
 => [distributor] resolving provenance for metadata file            0.0s
[+] up 14/14
 ✔ Image hw_content-worker2                Built                   58.4s
 ✔ Image hw_content-distributor            Built                   58.3s
 ✔ Image hw_content-collector              Built                   58.4s
 ✔ Image hw_content-sum_subscriber         Built                   58.4s
 ✔ Image hw_content-mut_subscriber         Built                   58.4s
 ✔ Image hw_content-div_subscriber         Built                   58.3s
 ✔ Image hw_content-worker1                Built                   58.3s
 ✔ Container hw_content-collector-1        Started                  0.7s
 ✔ Container hw_content-div_subscriber-1   Started                  0.5s
 ✔ Container hw_content-worker1-1          Started                  0.3s
 ✔ Container hw_content-worker2-1          Started                  0.3s
 ✔ Container hw_content-sum_subscriber-1   Started                  0.4s
 ✔ Container hw_content-mut_subscriber-1   Started                  0.4s
 ✔ Container hw_content-distributor-1      Started                  0.5s
```

```
worker2-1       |
distributor-1   | Sending task --- div (4 & 9) waiting for available worker to handle
distributor-1   | --- Sent ---
worker1-1       |
worker1-1       | WORKER-1: Handling the task
worker1-1       | [x] 4 / 9
worker1-1       | Result: 0.4444444444444444 --- Task has been processed by WORKER #1
worker1-1       | Data has been processed, sending them to the collector ...
worker1-1       | --- Sent ---
worker1-1       |
distributor-1   | Sending task --- sum (10 & 1) waiting for available worker to handle
distributor-1   | --- Sent ---
worker2-1       |
worker2-1       | WORKER-2: Handling the task
worker2-1       | [x] 10 + 1
worker2-1       | Result: 11 --- Task has been processed by WORKER#2
worker2-1       | Data has been processed, sending them to the collector ...
worker2-1       | --- Sent ---
worker2-1       |
distributor-1   | Sending task --- mut (9 & 19) waiting for available worker to handle
distributor-1   | --- Sent ---
worker1-1       |
worker1-1       | WORKER-1: Handling the task
worker1-1       | [x] 9 x 19
worker1-1       | Result: 171 --- Task has been processed by WORKER #1
worker1-1       | Data has been processed, sending them to the collector ...
worker1-1       | --- Sent ---
worker1-1       |
distributor-1   | Sending task --- mut (10 & 18) waiting for available worker to handle
distributor-1   | --- Sent ---
worker2-1       |
worker2-1       | WORKER-2: Handling the task
worker2-1       | [x] 10 x 18
worker2-1       | Result: 180 --- Task has been processed by WORKER#2
worker2-1       | Data has been processed, sending them to the collector ...
worker2-1       | --- Sent ---
worker2-1       |
distributor-1   | Sending task --- sum (15 & 5) waiting for available worker to handle
distributor-1   | --- Sent ---
worker1-1       |
worker1-1       | WORKER-1: Handling the task
worker1-1       | [x] 15 + 5
worker1-1       | Result: 20 --- Task has been processed by WORKER #1
worker1-1       | Data has been processed, sending them to the collector ...
worker1-1       | --- Sent ---
worker1-1       |
distributor-1   | Sending task --- mut (12 & 6) waiting for available worker to handle
distributor-1   | --- Sent ---
worker2-1       |
worker2-1       | WORKER-2: Handling the task
worker2-1       | [x] 12 x 6
worker2-1       | Result: 72 --- Task has been processed by WORKER#2
worker2-1       | Data has been processed, sending them to the collector ...
```

D. Kesimpulan

Berdasarkan hasil implementasi dan pengujian yang telah dilakukan. program ini berhasil menerapkan kombinasi pola komunikasi Push-Pull dan Publish-Subscribe menggunakan Python dengan library PyZMQ (ZeroMQ). Pada pola Push-Pull, Distributor berhasil mengirimkan tugas perhitungan kepada Worker 1 dan Worker 2. Kedua Worker dapat menerima dan memproses operasi penjumlahan, perkalian, serta pembagian, kemudian berhasil mengirimkan hasilnya ke Collector.

Selanjutnya, pada pola Publish-Subscribe. Collector berhasil mempublikasikan hasil perhitungan kepada Subscriber sesuai dengan topic SUM, MUT, dan DIV. Selain itu seluruh komponen program berhasil dijalankan menggunakan Docker Compose dan dapat saling berkomunikasi melalui jaringan antar-container.

Dengan demikian, implementasi ini menunjukkan bahwa kedua pola komunikasi dapat digabungkan untuk mendistribusikan tugas dan menyampaikan hasil perhitungan pada sistem terdistribusi.